

TEAM SUPERTRONICS

Analog Audio Effects Processor

EN2091 Analog Project

DATA SHEET

GENERAL DESCRIPTION

This product is a monolithic analog audio effects processor designed for high-fidelity audio signal manipulation. It integrates four distinct processing modules—Distortion, Equalization, Phasing, and Tremolo—into a single signal chain.

Powered by a linear regulated dual-rail power supply ($\pm 12\text{V}$), the device ensures high headroom and low noise floor suitable for instrument-level signals (Electric Guitar/Synthesizer). The design utilizes discrete analog components and operational amplifiers to maintain a warm, organic signal character.

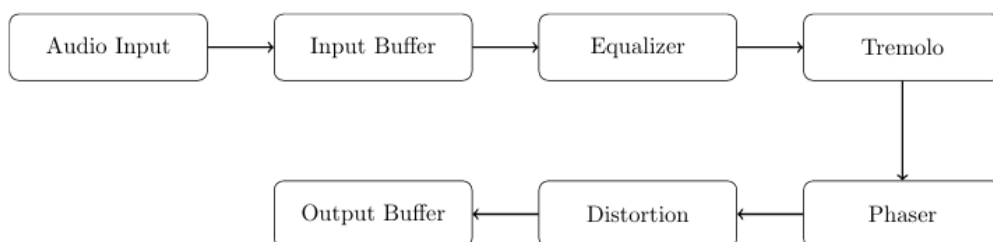
KEY FEATURES

- **3-Band Active EQ:** Independent control over Bass, Mid, and Treble frequencies.
- **Analog Tremolo:** LFO-driven amplitude modulation with independent Rate and Depth controls.
- **Multi-Stage Phaser:** 6-stage all-pass filter topology for deep sweeping effects.
- **Variable Distortion:** Wide gain range with diode-based clipping.
- **Linear Power Supply:** Transformer-isolated low-noise design using 7812/7912 regulators.
- **High Headroom:** $\pm 12\text{V}$ Dual Rail supply prevents unwanted clipping in clean stages.

APPLICATIONS

- Electric Guitar Effects Units
- Analog Synthesizer Processing
- Educational Signal Processing Demonstrations

SYSTEM BLOCK DIAGRAM



ELECTRICAL CHARACTERISTICS

Unless otherwise noted: $T_A = 25^\circ\text{C}$, $V_{IN} = 230\text{V AC}$, $f_{in} = 1\text{kHz}$.

Parameter	Min	Typ	Max	Unit
Power Supply				
Mains Input Voltage	220	230	240	V_{AC}
Internal DC Rails	-	± 12	-	V_{DC}
Quiescent Current (I_Q)	-	76	-	mA
Max Current Draw	-	-	200	mA
Transformer Rating	-	1	-	A
Fuse Rating	-	500	-	mA
Audio Performance				
Input Impedance	-	1	-	$M\Omega$
Output Impedance	-	1	-	$k\Omega$
Frequency Response	20	-	20k	Hz
Signal-to-Noise Ratio	-	> 80	-	dB
THD + N	-	< 0.05	-	%
Equalizer Characteristics				
Bass Adjust (100Hz)	-15	0	+15	dB
Mid Adjust (1kHz)	-12	0	+12	dB
Treble Adjust (4kHz)	-15	0	+15	dB

MODULE DESCRIPTION

This project consists of five distinct circuit modules, each responsible for a specific stage of signal processing. The following sections detail the design and functionality of the Power Supply, Equalizer, Tremolo, Phaser, and Distortion stages.

Power Supply

The power supply module is critical for ensuring low-noise operation across all audio stages. It is designed to convert the input voltage into a stable DC supply required by the active components. This circuit utilizes voltage regulation to eliminate ripple and ensures a constant voltage level regardless of load fluctuations. Additionally, a voltage divider network creates a virtual ground reference (V_{ref}), allowing single-supply op-amps to process AC audio signals effectively.

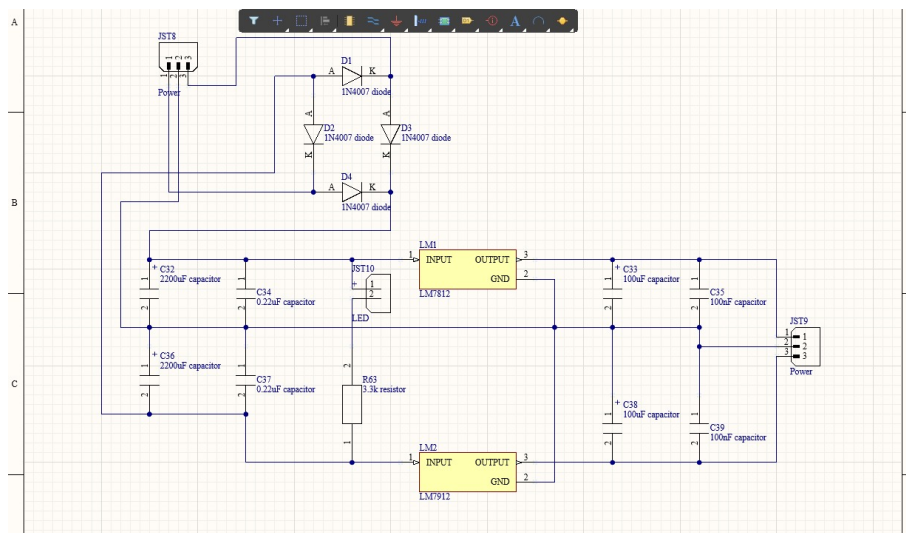


Figure 1: Power Supply Circuit Schematic showing regulation and filtering stages.

Input Buffer and Equalizer

The first stage of the signal path features a high-impedance input buffer. This ensures that the input signal is not loaded down, preserving high-frequency content and signal integrity before processing. Following the buffer, an active equalizer circuit allows for tone shaping. This stage employs an op-amp based filter topology to boost or attenuate specific frequency bands, allowing the user to tailor the frequency response prior to the modulation and distortion effects.

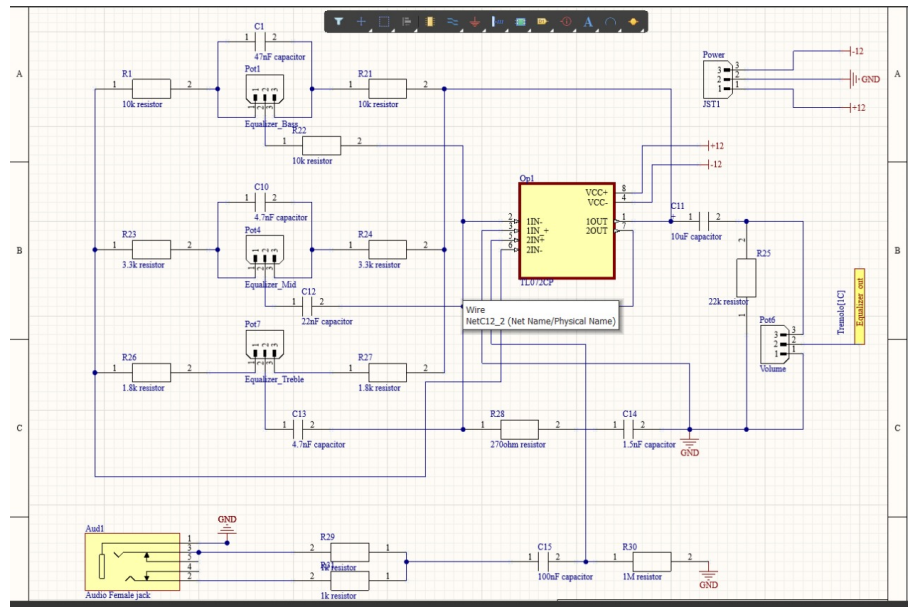


Figure 2: Input Buffer and Equalizer Circuit Schematic.

Tremolo

The Tremolo module produces amplitude modulation to create a rhythmic pulsing effect. The core of this circuit is a Low-Frequency Oscillator (LFO) which generates a triangle control waveform. This control signal interacts with an optocoupler to vary the volume of the audio signal cyclically. The circuit includes controls for 'Rate' (speed of the LFO) and 'Depth' (intensity of the volume reduction).

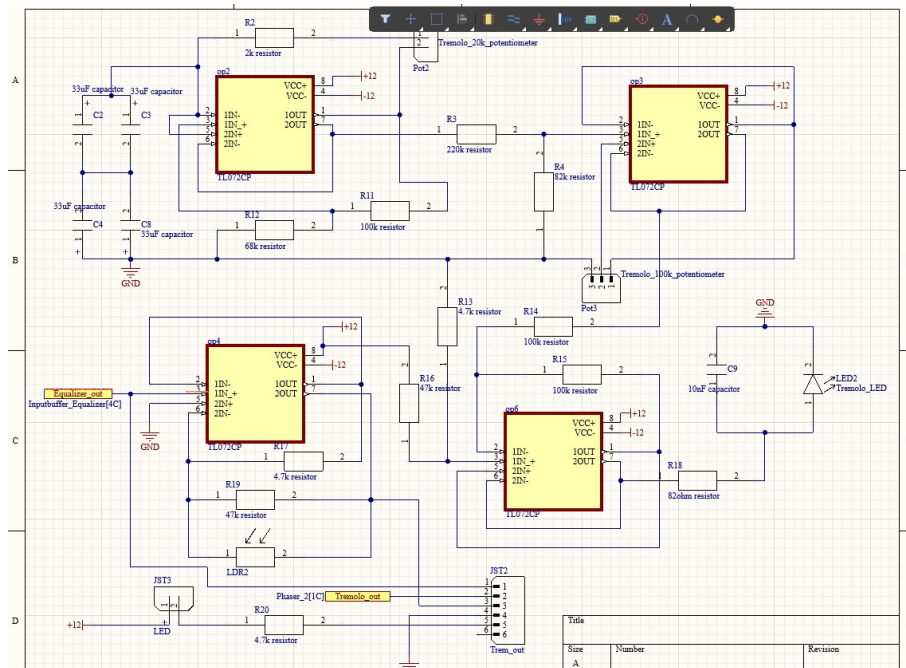


Figure 3: Tremolo Circuit Schematic showing LFO and modulation path.

Phaser

The Phaser module creates a sweeping comb-filter effect by splitting the audio signal into two paths. One path remains dry, while the other passes through a series of all-pass filters that shift the phase of the signal. When the phase-shifted signal is mixed back with the dry signal, cancellations occur at specific frequencies. The phase shift is modulated by an LFO, causing the frequency notches to sweep up and down the spectrum.

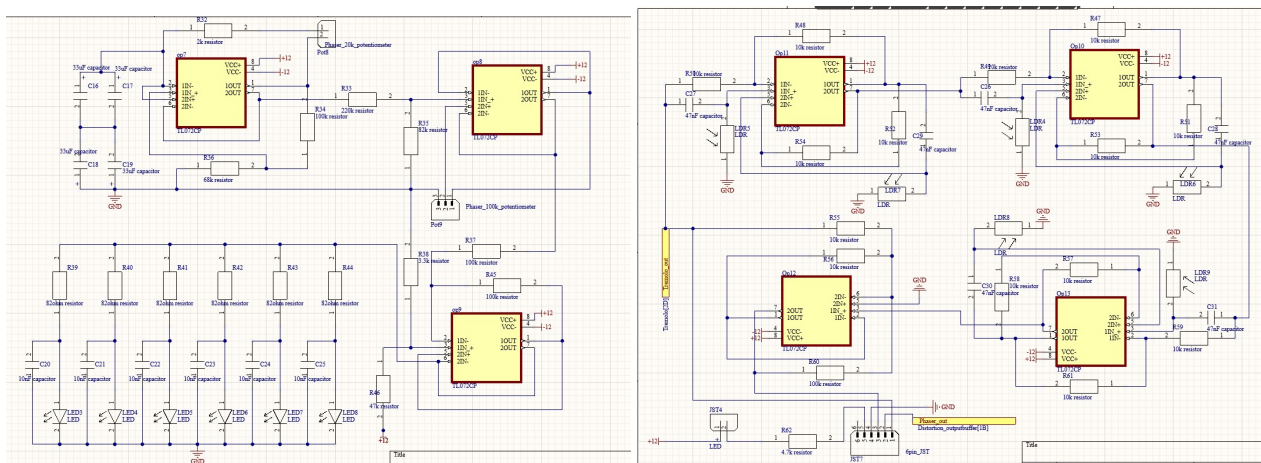


Figure 4: Phaser Circuit Schematics: Phase-Shifting Stages (Left) and Control Logic (Right).

Distortion and Output Buffer

The final module is the Distortion circuit, designed to add sustain and harmonic saturation to the signal. This is achieved by applying high gain to the audio signal and driving it into a clipping stage. The clipping mechanism utilizes a specific arrangement of diodes to limit the signal amplitude, squaring off the waveform tops and bottoms. This introduces odd-order harmonics, resulting in the characteristic "crunch" or "fuzz" tone associated with hard rock and metal music.

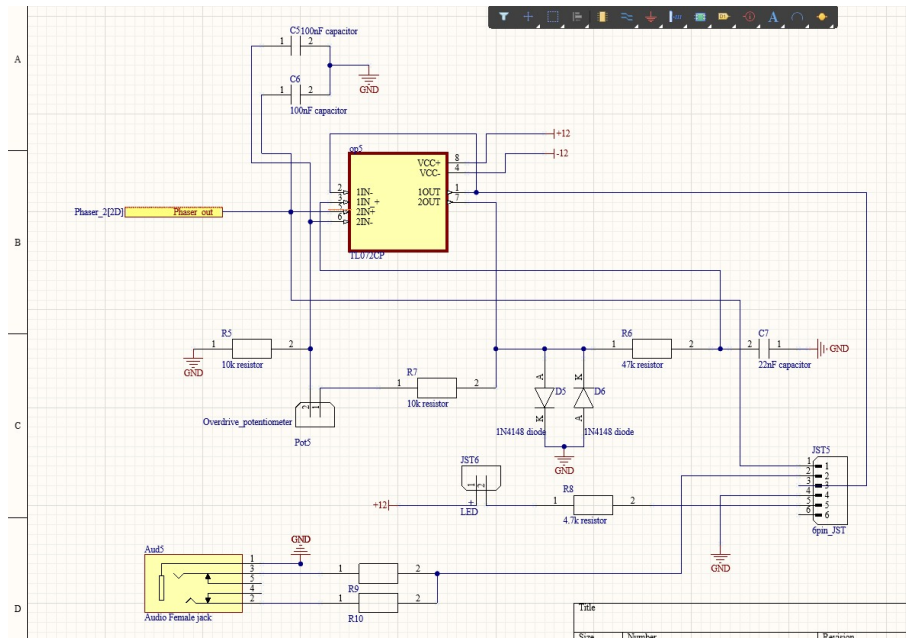


Figure 5: Distortion Circuit Schematic showing gain and clipping stages.

PIN CONFIGURATION

Input Jack: 3.5mm AUX Stereo Jack

Output Jack: 3.5mm AUX Stereo Jack

CONTACT / SUPPORT

For further technical details or schematics, please contact the design team at:
supertronics@gmail.com